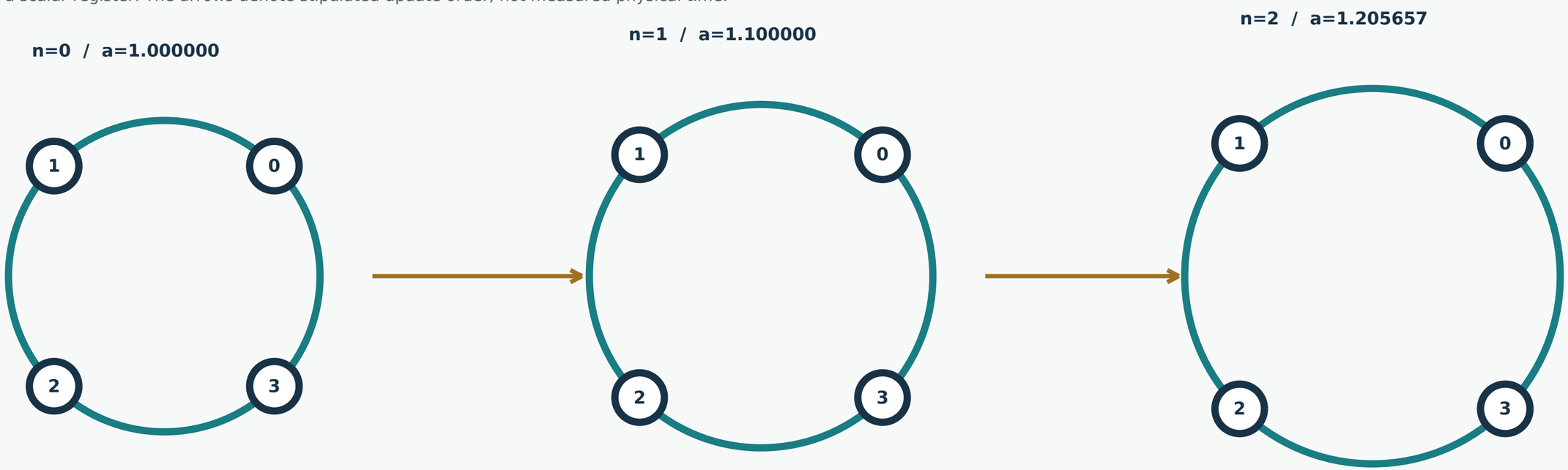
Toy Universe Map

01 / Finite layout: what the stipulated world contains, not a map of observed spacetime

A FOUR-SITE RING THROUGH THREE FORMAL UPDATES

Size a is a scalar register. The arrows denote stipulated update order, not measured physical time.



Formal slice 0: same 4 labels

Formal slice 1: same 4 labels

Formal slice 2: same 4 labels

CORE BENCHMARK VALUES

Matter state

A density matrix on C^(2^4): 16 basis configurations; complex amplitudes are installed, not derived.

Record outcomes

Each site: silent, 0, or 1. Up to 3^4 = 81 complete outcome labels per step.

Geometry register

a > 0; scalar size grows by a chosen recurrence using matter occupation q.

Not depicted

No calibrated distance, Lorentz cone, 3+1D spacetime or physical apparatus.

Dependency and Import Map

02 / Solid: stipulated/verified mathematical dependency. Dashed: missing physical bridge.

AXIOM LAYER (ALL IMPORTED)

Cyclic site set

N finite sites, adjacency

Complex matter space

H = tensor C2, rho >= 0

Record instrument

Kraus operators + trace law

Formal update index

n, delta; not a clock

Scalar size register

a > 0, Omega parameters

Matter dynamics

H_m(a) = -(J/a) sum ZZ - h sum X; U_a = exp(-i delta H_m(a)). Unitarity is proved from Hermiticity.

Record channel

rho(next) = sum_r M_r U rho U* M_r*. The Kraus completeness identity proves trace preservation and positivity.

parallel step

Scalar expansion

a(next) = a + delta a sqrt(F(a,q)); q uses PRE-STEP rho. This updates in parallel with the record channel. GR is not obtained.

MATHEMATICAL OUTPUTS (CONDITIONAL, CHECKED)

Interference / dephasing

Single-site phase fringes: p0 = [1 + (1-kappa) cos(phi)]/2. Analytical proof and numerical verification.

Classical probability limit

With full basis dephasing, diagonal probabilities follow the unistochastic transition |U_ji|^2.

Positive toy expansion

A four-site trajectory was recomputed. The scalar recurrence is internally well defined for finite steps.

PHYSICAL IDENTIFICATION: NO APPROVED BRIDGE

No independently selected Born law, actual clock, metric, stress-energy, calibration or physical emergence.

Candidate Adversarial Screen

03 / Four testbeds compared; locally completed reviews do not imply parent-phase advancement

MODEL	WHAT IT CAN SHOW	WHAT IT CANNOT SELECT	DISPOSITION
CLS / finite stochastic	Stationary laws, discrete records	Exact unrestricted phase interference under only affine classical probabilities	Comparator retained
CTX / withdrawn TO-0	Contextual and nonsignalling tables	Classical vs quantum vs PR physical theory	Ontological candidacy withdrawn
ORD / bare causal order	Partial order, adjacency and counting	Probability, physical clock or unique continuum metric	No preferred candidate
FAR-1 / complete toy	Finite amplitude dynamics, records, positive size scalar	Why its imported laws physically exist	Stipulated object only

FOCUSED LOCAL SUB-PCP CHECKPOINTS

TU-01
Selection: negative

TU-02
Finite proof: pass

TU-03
Classical mechanics: open

TU-04
Cosmology / prizes: open

TU-05
QA and evidence: scoped

Cosmology: Obligation Atlas

04 / These are routes requiring additional physics and data, not claimed explanations

EXPANSION & DISTANCES

Toy: positive a(n). Missing: FRW constraints, clocks, standard candles/rulers and multi-survey likelihood.

DARK ENERGY

Toy: chosen positive constant term. Missing: warranted equation of state, independent data comparison, uncertainty.

DARK MATTER & STRUCTURE

Toy: no halos or lensing. Missing: gravitational perturbations, nonlinear growth, particle or modified-gravity tests.

INFLATION & INITIAL STATE

Toy: arbitrarily chosen rho(0). Missing: primordial spectrum, mode dynamics, reheating and discriminating predictions.

CMB, BAO, NUCLEOSYNTHESIS

Toy: no photon-baryon plasma. Missing: calibrated particle content, transfer functions, independent datasets.

BARYON ASYMMETRY & PARTICLES

Toy: site labels only. Missing: field content, charges, symmetries, collisions, neutrinos and matter-antimatter mechanism.

BLACK HOLES & ENTROPY

Toy: no horizon or metric. Missing: relativistic gravity, causal boundary, consistent quantum fields and information balance.

SINGULARITIES & COSMIC FATE

Toy: finite positive recurrence. Missing: global extension/singularity theorem and boundary physical criterion.

HUBBLE / GROWTH TENSIONS

Toy: no dimensional H0 or growth factor. Missing: preregistered datasets, nuisance covariance, independent calibration.

GRAVITATIONAL WAVES & TESTS

Toy: no propagating geometry. Missing: Lorentzian gravitational dynamics and precise strong-field/solar-system tests.

External comparators: Particle Data Group 2025 cosmology and DESI DR2 July 2026. No fitting was performed.

Mathematical Boundary Atlas

05 / Pure-mathematical problems are not consequences of the toy geometry

P vs NP

No uniform complexity-class separation follows from finite toy simulation.

Riemann hypothesis

No result about all nontrivial zeta zeros follows from a finite spectrum.

Birch and Swinnerton-Dyer

No theorem links arbitrary elliptic-curve rank and L-function behavior.

Hodge conjecture

No theorem on rational Hodge classes / algebraic cycles is supplied.

Yang-Mills mass gap

A finite spin energy gap is not the constructive 4D continuum Yang-Mills theorem.

Navier-Stokes existence/smoothness

No 3D continuum theorem; Clay lists Active while evaluating September 2026 announcement.

Poincare conjecture

Already solved by Perelman. No new result from FAR-1.

PI (DEFINED, NOT EMERGENT)

pi = 4 arctan(1) = 3.141592653589... Imported real mathematics.

TAU-TURN (DEFINED, NOT EMERGENT)

tau_turn = 2 pi = 6.283185307179... Not a prephysical frequency or signal.

Evidence and Review Control

06 / What was tested, what is missing, and why release validity is not scientific proof

L0 / STIPULATED Hilbert space, Born-style record rule, formal n, scalar expansion are axioms.	**BOUNDARY / REJECTION** No proof of physical origin.
L1 / PROVED CONDITIONAL Finite trace/positivity and record normalization follow from Kraus completeness.	**BOUNDARY / REJECTION** Proved only given L0.
L2 / COMPUTED Interference, 4-site trajectory, 28-round Bell-control regression numerical values.	**BOUNDARY / REJECTION** Known numerical tolerance; not data.
L3 / PHYSICAL BRIDGE Actual preparation, detector, time, geometry, calibration, causal independence.	**BOUNDARY / REJECTION** Missing.
L4 / FOUNDATION / COSMOLOGY GR, real universe, quantum/classical emergence, Millennium problem proof.	**BOUNDARY / REJECTION** Not established.

Source integrity is not proof certification; local sub-PCP completion is not registered parent admission.

Provenance, References, Closure

07 / Session artifact closed; parent Initial External Grounding remains OPEN

GOVERNING RESEARCH LINEAGE

Frozen baseline: 10 November 2026; SHA-256 fde41b3789930b0848cffd80b0ced68b052645808c393525266b1457d6d669cf. Updated PCP 1.0: Core R13; Mathematics R5; Science R11; Statistics White R2; Quality Assurance Lime R3. Director B Bell Memory/Sequential QA is internal research provenance, not independent empirical replication.

E1 / QUANTUM LATTICES

Arrighi 2019 QCA overview; Farrelly 2020 review of quantum cellular automata.

E2 / DECOHERENCE

Stanford Encyclopedia of Philosophy, Role of Decoherence; Consistent Histories.

E3 / SPACETIME

Surya 2019 causal sets; Discrete Approaches to Quantum Gravity in Four Dimensions.

E4 / REAL COSMOLOGY

Particle Data Group 2025 Astrophysics & Cosmology; DESI DR2 30 July 2026.

E5 / MATH PROBLEM STATUS

Clay Mathematics Institute seven prize problems and 11 September 2026 Navier-Stokes notice.

E6 / MODEL SELECTION

Abramsky and Brandenburger 2011 contextuality; NIST metrological traceability.

FINAL RESEARCH DISPOSITION

FAR-1 is a fully stipulated, finite, internally tested mathematical universe. It is NOT a physically selected total ontology. TO-0 remains withdrawn; no successor is warranted. Baseline proof gaps, physical observation bridges, and comparative PCP effectiveness remain open. Parent Unary PCP: Initial External Grounding, OPEN.